

TURPAN, China

WMO: 515730

Lat: 42.9503N

Lon: 89.2307E

Elev: 129

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.6	(c) 10.5	(d) -9.0	(e) 3.4	(f) 26.8	(g) -6.1	(h) 4.0	(i) 22.2	(j) 7.6	(k) 22.6	(l) 6.7	(m) 21.5	(n) 1.0	(o) 40	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	108.5	70.3	105.8	69.6	103.3	69.0	73.4	100.2	72.0	98.5	70.8	97.3	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	93.7	85.5	62.8	85.8	86.4	60.6	79.3	87.8	37.2	100.2	35.9	98.4	34.9	97.2	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	9.4	7.5	7.0	114.0	3.6	2.7	4.3	116.0	2.2	117.6	0.2	119.1	-2.5	121.1
			5.4	77.1	3.3	3.8	3.1	79.8	1.2	82.1	-0.7	84.2	-3.0	86.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	20.9	35.6	53.3	69.3	80.2	89.3	92.3	89.3	78.0	60.6	42.0	24.9
	DBStd	25.57	5.65	7.18	8.81	7.90	6.31	5.48	4.97	5.23	6.59	7.65	7.37	6.35
	HDD50	2409	903	404	66	1	0	0	0	0	0	7	251	777
	HDD65	4729	1368	823	373	48	1	0	0	0	3	182	689	1242
	CDD50	6579	0	1	170	579	937	1180	1310	1217	839	334	12	0
	CDD65	3423	0	0	11	176	473	730	845	752	392	44	0	0
	CDH74	47796	0	0	94	1769	5764	11062	13394	11128	4263	322	0	0
	CDH80	29291	0	0	11	673	3048	7167	9115	7137	2071	69	0	0
	WSAvg	2.6	1.7	2.1	2.8	3.4	3.6	3.5	3.1	2.7	2.0	1.6	1.6	1.6
	PrecAvg	0.6	0.0	0.0	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0
(15) Precipitation	PrecMax	1.5	0.4	0.2	1.0	0.1	0.4	0.7	0.6	0.4	0.5	0.4	0.5	0.3
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.4	0.1	0.0	0.2	0.0	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1
	PrecStd	0.4	0.1	0.0	0.2	0.0	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	37.3	57.4	81.3	95.5	103.4	110.1	113.2	110.2	101.1	86.3	64.5	43.0	35.4
	0.4% MCWB	29.4	41.8	54.2	61.5	65.9	69.3	71.5	70.3	67.4	60.2	48.2	35.4	35.4
	2% DB	33.8	53.1	76.4	90.8	99.8	106.7	109.5	106.7	97.2	81.4	60.3	38.9	31.9
	2% MCWB	27.2	39.3	51.9	59.7	64.8	68.7	71.4	69.8	66.0	58.2	45.9	31.9	31.9
	5% DB	31.3	50.2	72.4	87.2	96.7	104.2	106.5	103.9	93.9	77.5	56.8	35.9	29.9
	5% MCWB	25.5	37.5	50.1	58.2	64.0	68.1	70.5	69.4	64.9	56.3	44.1	29.9	29.9
	10% DB	29.1	47.1	67.9	83.7	93.3	101.4	103.8	101.0	90.6	73.6	53.4	33.4	28.2
	10% MCWB	24.0	35.6	47.9	56.6	62.8	67.6	70.0	68.8	63.5	54.6	42.3	28.2	28.2
	0.4% WB	30.2	42.4	55.1	62.8	69.1	72.9	76.5	74.6	70.7	61.5	49.4	35.9	35.9
	0.4% MCDB	36.0	56.1	79.0	91.0	97.1	99.8	101.9	98.0	94.0	82.6	62.1	42.3	42.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	27.6	39.7	52.7	60.7	66.6	71.3	74.1	72.7	68.2	59.3	46.9	32.7	32.7
	2% MCDB	33.1	52.0	74.7	88.6	95.5	97.9	101.2	97.7	91.8	78.7	58.3	37.6	37.6
	5% WB	25.9	37.8	50.4	59.0	65.1	70.0	72.6	71.4	66.2	57.3	44.8	30.6	30.6
	5% MCDB	31.0	49.4	71.4	85.1	93.4	97.2	100.1	97.2	89.3	75.2	55.6	35.6	35.6
	10% WB	24.1	35.8	48.2	57.3	63.7	68.9	71.4	70.2	64.6	55.4	42.9	28.2	28.2
	10% MCDB	28.9	46.5	67.2	81.6	90.9	96.4	98.3	96.3	88.1	72.1	52.6	33.4	33.4
(35) Mean Daily Temperature Range	MDBR	12.5	16.6	19.3	21.5	21.9	21.6	21.2	22.6	23.2	21.9	16.3	10.9	10.9
	5% DB	13.9	18.7	22.6	25.9	26.7	25.0	25.5	26.2	25.7	25.0	20.6	12.1	12.1
	5% WB	9.9	10.9	10.6	10.5	10.3	8.5	8.5	8.6	9.2	9.8	10.5	8.1	8.1
	MCDBR	13.5	18.5	21.8	24.2	24.1	22.2	22.3	23.5	24.2	24.6	20.3	11.4	11.4
	MCWBR	9.7	10.9	10.3	9.8	9.8	8.3	8.6	8.8	9.3	9.6	10.5	7.9	7.9
	MCWBR	9.7	10.9	10.3	9.8	9.8	8.3	8.6	8.8	9.3	9.6	10.5	7.9	7.9
(40) Clear-Sky Solar Irradiance	taub	0.418	0.473	0.540	0.585	0.583	0.588	0.564	0.545	0.527	0.503	0.465	0.420	0.420
	taud	1.917	1.827	1.704	1.667	1.719	1.779	1.868	1.890	1.859	1.841	1.883	1.930	1.930
	Ebn at Noon	206	215	218	221	227	227	231	230	220	203	187	190	190
	Edh at Noon	40	52	67	75	73	69	62	59	56	51	42	37	37
(44) All-Sky Solar Radiation	RadAvg	711	1032	1447	1825	2124	2175	2067	1923	1658	1201	790	611	611
	RadStd	40	51	62	115	88	137	103	109	66	55	44	42	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.46	N/A	N/A	+1.87	-1.04	-2.88	N/A	N/A	+536	+394
Regional (3 neighbors)	+0.85	N/A	N/A	N/A	N/A	-1.49	N/A	N/A	+236	+162

Nomenclature: See separate page